

Jiaheng Chen

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Education

University of Chicago

Ph.D. Student

Chicago, US

2022 – Now

- Computational and Applied Mathematics
- Advisor: Daniel Sanz-Alonso

Shanghai Jiao Tong University

Undergraduate Student

Shanghai, China

2018 – 2022

- Mathematics and Applied Mathematics, Zhiyuan College

Research Interests

I am broadly interested in the mathematical foundations of data science, scientific machine learning, operator learning, high-dimensional probability and statistics, and Monte Carlo methods.

Publications

- [1] High-dimensional quasi-Monte Carlo via combinatorial discrepancy. With H. Jiang and N. Kirk. *Journal of Complexity*, 96, 102053, 1-23, (2026).
- [2] Sharp concentration of simple random tensors II: asymmetry. With D. Sanz-Alonso. *Information and Inference*, 15(2), 1-46, (2026).
- [3] On the estimation of Gaussian moment tensors. With O. Al-Ghattas and D. Sanz-Alonso. *Electronic Communications in Probability*, 30, 1-15, (2025).
- [4] Sharp concentration of simple random tensors. With O. Al-Ghattas and D. Sanz-Alonso. *Information and Inference*, 14(4), 1-41, (2025).
- [5] Precision and Cholesky factor estimation for Gaussian processes. With D. Sanz-Alonso. *SIAM/ASA Journal on Uncertainty Quantification*, 13(3), 1085-1115, (2025).
- [6] Covariance operator estimation: sparsity, lengthscale, and ensemble Kalman filters. With O. Al-Ghattas, D. Sanz-Alonso and N. Waniorek. *Bernoulli*, 31(3), 2377-2402, (2025).
- [7] A machine learning framework for geodesics under spherical Wasserstein-Fisher-Rao metric and its application for weighted sample generation. With Y. Jing, L. Li and J. Lu. *Journal of Scientific Computing*, 98(5), 1-34, (2024).

Preprints

- [1] Sharp Frobenius-norm concentration for sample moment tensors. With D. Sanz-Alonso. *arXiv*: 2608.11084
- [2] Optimal multiscale learning of linear operators. With D. Sanz-Alonso. *arXiv*: 2606.16913

- [3] Concentration inequalities for sample cross-covariances. With D. Sanz-Alonso.
arXiv: 2605.16733
- [4] Convergence rates for learning pseudo-differential operators. With D. Sanz-Alonso.
arXiv: 2601.04473
- [5] Optimal estimation of structured covariance operators. With O. Al-Ghattas, D. Sanz-Alonso, and N. Waniorek. *arXiv*: 2408.02109
- [6] Fluctuation suppression and enhancement in interacting particle systems. With L. Li.
arXiv: 2204.07757

Fellowships and Awards

- William Rainey Harper Dissertation Fellowship, University of Chicago 2026
Awarded in recognition of a PhD candidate's record of achievement and professional promise; one of the University of Chicago's highest honors
- Travel Award, SIAM Conference on Uncertainty Quantification 2026
- Neubauer Family Distinguished Doctoral Fellowship, University of Chicago 2022 – 2027
- Zhiyuan Outstanding Student Scholarship, Shanghai Jiao Tong University 2022
- Excellent Bachelor's Thesis Award, Shanghai Jiao Tong University 2022
- Shanghai Outstanding Graduate 2022
- Outstanding Winner, SIAM Award, COMAP Scholarship Award 2021
Mathematical Contest in Modeling and Interdisciplinary Contest in Modeling
- National Scholarship, Ministry of Education of China 2020
- Zhiyuan Honors Scholarship, Shanghai Jiao Tong University 2018 – 2022

Conferences and Seminars

- SIAM Conference on Computational Science and Engineering, Pittsburgh, US Feb 22 – 26, 2027
Minisymposium: "Probability models and distributional data in scientific machine learning"
- SIAM Conference on Mathematics of Data Science, Salt Lake City, US Nov 16 – 20, 2026
Minisymposium: "Recent Advances in Operator Learning: Theory, Methods, and Applications"
- SIAM Annual Meeting, Cleveland, US Jul 6 – 10, 2026
Minisymposium: "From Theory to Computation: Modern Methods for PDE Inverse Problems"
- SIAM Conference on Uncertainty Quantification, Minneapolis, US Mar 22 – 25, 2026
Minisymposium: "Recent Advancements in Operator Estimation and Learning"
- CAM & Statistics Student Seminar, University of Chicago Mar 4, 2026
- Joint Statistical Meetings (JSM), Nashville, US Aug 2 – 7, 2025
Section: "Functional and Shape-Constrained Data Analysis: Methods and Applications"
- DACO Seminar, Department of Mathematics, ETH Zurich (*virtual*) May 22, 2025
- Stat & CAM Student Research Poster Day, University of Chicago May 20, 2025
- Workshop "Statistics Meets Tensors: Methodology, Theory, and Applications", Institute for Mathematical and Statistical Innovation (IMSI) May 5 – 9, 2025

- Reading Seminar on Stochastic Many-body Systems (*virtual*) Apr 2, 2025
- Workshop “Kernel Methods in Uncertainty Quantification and Experimental Design”, Institute for Mathematical and Statistical Innovation (IMSI) Mar 31 – Apr 4, 2025
- Institute of Natural Sciences, Shanghai Jiao Tong University Sep 1, 2024
- SIAM Conference on Uncertainty Quantification, Trieste, Italy Feb 27 – Mar 1, 2024
Minisymposium: “Statistical and Data-Assisted Modeling Approaches for Forecasting and Uncertainty Quantification of Complex Multiscale Systems in Real-World Applications”
- CAM & Statistics Student Seminar, University of Chicago Oct 10, 2023

Conference Organization

- SIAM Conference on Computational Science and Engineering, Pittsburgh, US Feb 22 – 26, 2027
Co-organizer, minisymposium “Error Analysis for Operator Learning”, with Margaret Trautner
- SIAM Conference on Uncertainty Quantification, Minneapolis, US Mar 22 – 25, 2026
Co-organizer, minisymposium “Recent Advancements in Operator Estimation and Learning”, with Omar Al-Ghattas and Nathan Waniorek

Teaching

Teaching Assistant at University of Chicago

- STAT/CAAM 31511: Monte Carlo Simulation Spring 2026
- STAT/CAAM 31521: Applied Stochastic Processes Spring 2025
- STAT/CAAM 31050: Applied Approximation Theory Spring 2024
- STAT/CAAM 38100: Measure-Theoretic Probability I Winter 2024
- STAT/CAAM 31150: Inverse Problems and Data Assimilation Autumn 2023

Referee Service

Advances in Computational Mathematics, Biostatistics